TECHNICAL ENGINEERING SPECIFICATION

GCMS Clinical AI Backend Architecture

Technical REST API & System Architecture Specification

System Name:	GCMS Clinical AI Consultation Backend
Document Type:	Technical Architecture & REST API Specification
Edition:	FastAPI + Ollama Production Reference v1.2
Engineering Leads:	Senior AI Backend Engineer & Healthcare Architect
Technology Stack:	Python 3.11+, FastAPI, SQLAlchemy 2.0, PostgreSQL 16, Ollama
Publication Date:	November 2026

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Complies with Singapore MOH, HSA, and PDPA Healthcare Guidelines

1. System Architecture Overview

The GCMS Clinical AI Backend is built as an asynchronous Backend-for-Frontend (BFF) leveraging FastAPI and SQLAlchemy 2.0. It acts as a secure mediation layer between the clinician’s React frontend, the existing Clinic Management System (CMS), the local Ollama AI inference runtime, and high-performance PostgreSQL terminology databases.

> **[IMPORTANT] Three-Tier Architectural Separation**
> Layer 1 (AI Reasoning): Ollama synthesizes unstructured consultation context into structured candidate suggestions. Layer 2 (Terminology Grounding): PostgreSQL FTS & Trigram engines verify SNOMED CT and SDD codes. Layer 3 (Deterministic Safety): Independent deterministic Python rules verify allergies, interactions, and contraindications.

2. REST API Specification (The 4 Core Services)

Endpoint Route	Method	Request Payload	Response & Responsibilities
/api/v1/terminology/snomed/search	GET	q={term}&page=1&page_size=20	Full-text (tsvector) + Trigram (pg_trgm) search on SNOM
/api/v1/terminology/sdd/search	GET	q={term}&page=1&page_size=20	Multi-field search on brand, generic, strength, HSA reg
/api/v1/ai/diagnoses/suggest	POST	{consultation_context, max_suggestions}	Differential diagnoses with supporting factors & SNOMED
/api/v1/ai/medications/suggest	POST	{selected_diagnosis, consultation_context}	Guideline therapeutics grounded in SDD with independent
/api/v1/health	GET	None	Service readiness probe testing PostgreSQL and Ollama c

3. Local Ollama AI Runtime Integration

All generative AI inference runs on locally hosted instances via Ollama. The integration layer features:

* Connection Pooling ? Persistent httpx.AsyncClient with keepalive connections and configurable timeouts (default 120s).
* Resilience & Retries ? Tenacity exponential backoff retries on transient connection dropouts.
* Strict JSON Mode ? Forces Ollama to emit structured JSON complying with Pydantic schemas, eliminating parsing failures.
* Zero Hallucinations ? The AI system prompts explicitly forbid inventing SNOMED CT or SDD codes. Terminology mapping is exclusively performed by the database ground-truth engine.

4. Database Architecture & PostgreSQL Schemas

Table Name	Primary Key	Indexed Search Strategy	Business Entity
snomed_concepts	concept_id (String 20)	B-Tree on active status	Core SNOMED CT concept registry
snomed_descriptions	description_id (String 20)	GIN on search_vector, pg_trgm on term	FSN, preferred terms, and clinical synonyms
sdd_medications	medication_id (String 50)	GIN tsvector, pg_trgm on generic/brand	HSA-registered Singapore medication catalogue
sdd_active_ingredients	id (UUID)	pg_trgm on ingredient_name	Chemical active ingredients and strengths
audit_events	id (UUID)	B-Tree on request_id, event_type, created_at	Immutable audit records of clinical AI operations
ai_request_logs	id (UUID)	B-Tree on request_id, created_at	Telemetry (tokens, latency, prompt versions)

5. Privacy Boundary & PII Scrubbing Engine

To enforce Section 5 of the architecture specification, all free-text notes pass through the PIIFilter before prompt construction:

* Singapore NRIC / FIN ? Regex pattern matches and redacts national identifiers [STFGM]\d{7}[A-Z] -> [NRIC_REDACTED].
* Contact Information ? Singapore phone numbers (+65 6/8/9xxx xxxx) and emails are redacted.
* Context Minimization ? ContextMinimizer strips patient identity fields (names, addresses, MRNs) prior to processing.

6. Independent Clinical Safety Engine

Medication candidates generated by Ollama are cross-checked against four independent safety algorithms:

* Allergy Cross-Reactivity ? Detects documented drug allergies and class cross-sensitivities (Penicillins, Cephalosporins, NSAIDs, Sulfas).
* High-Risk Drug Interactions ? Evaluates concurrent medications for major clinical interactions (e.g., Warfarin + NSAIDs, Simvastatin + Macrolides).
* Contraindication Rules ? Evaluates pregnancy teratogens (Category X/D) and renal clearance thresholds (eGFR < 30 mL/min for Metformin).
* Duplicate Therapy ? Flags simultaneous prescriptions sharing active chemical substances.